Lagrangian Branch-and-Bound for N-Track Railway Timetabling, with Compressed-Response Branch Screening: Implementation and an Honest Evaluation

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(working draft — N-track results, companion to the single-track paper)

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Abstract

We report a faithful, portable reimplementation of the Meng–Zhou (2014) N-track space-time Lagrangian relaxation (LR) for railway timetabling, wrap it in an exact LR-based branch-and-bound (B&B) using the train-segment *enumeration* branching of Zhou and Zhong (2005), accelerate the per-train shortest path with an exact heap-free dynamic program, and then ask — with deliberate honesty — whether low-rank *primal/dual response screening* (SVD and tensor) can choose branches as well as strong branching. Four findings. **(i)** The clean LR reproduces the published bounds on RAS Data Set 3 (lower bound $1785 \to 2279$, upper bound 3803, gap $\approx 40\%$). **(ii)** The LR-based B&B is exact: on a hand-checkable instance it proves the optimum (gap 0), and the segment-enumeration split (commit one train's segment, let LR offset the others) proves it in 19 nodes versus 51 for a single-cell split; the branch-*selection* criterion is a free, swappable heuristic that changes only tree size. **(iii)** The time-expanded graph is a DAG, so a single increasing-time sweep replaces Dijkstra and cuts the root LR from 7.31 s to 2.48 s ($\sim 2.9\times$) with identical shortest-path costs; an *approximate* path is inadmissible because it would invalidate the LR lower bound. **(iv)** Branch screening: at a single root node, an SVD compressed-response score over the primal flow (resp. dual prices) contains the strong-branching choice perfectly (Containment@1= 1, resp. rank correlation +0.70); *but this does not replicate across nodes* — over six B&B nodes no unsupervised SVD or tensor score beats a random or pseudo-cost baseline, mirroring the single-track result that only a *learned* screen transfers. Finally, we show the optimality gap is limited by *bound tightness, not compute*: faster nodes do not lift the global lower bound above the root LR value, which motivates a stronger relaxation (cuts, or an ADMM consensus bound) rather than more node throughput.

1 Introduction

Railway timetabling assigns each train a conflict-free trajectory over shared time–space resources (track segments, sidings, junctions) subject to running times, dwell, meet–pass order, and link capacity. The Meng–Zhou (2014) formulation casts this as a cumulative-flow space-time network and dualizes the link-capacity constraints, decomposing the problem into per-train time-dependent shortest paths priced by Lagrangian multipliers. The relaxation is strong and cheap, but it leaves an integrality (duality) gap; closing it requires branch-and-bound, whose cost is dominated by two things: the per-node bound (many shortest paths) and the choice of *which* conflict to branch on.

This paper has two aims, and we are explicit that the second produces a partly negative result. The first is engineering: build a clean, portable, exactness-preserving LR-based B&B that reproduces the literature and actually runs on the RAS instances. The second is scientific: test whether the low-rank structure of the solver's *response field* — the primal flows and dual prices it emits at each node — can screen branching candidates, so that expensive strong-branching child solves are spent only where they matter. We test primal screening, dual screening, their SVD compressions, and a tensor (CP/HOSVD) fusion of the two, against full strong branching, on real RAS data.

Contributions.

1. A standard-C++17, MFC-free reimplementation of the Meng–Zhou FastTrain LR that reproduces the published lower/upper bounds (§2).
2. An exact LR-based B&B with per-train time-windowed restrictions, warm-started node bounds, segment-enumeration branching (Zhou–Zhong 2005), best-first search, and a *separation of the sound disjunction from the heuristic selection criterion* (§3).
3. A heap-free exact DAG shortest path with a $\sim$2.9$\times$ speedup and a precise statement of why approximate paths are inadmissible (§4).
4. An honest primal/dual/SVD/tensor branch-screening study: a striking single-node result that *fails to replicate* across nodes, and the diagnosis that the robust screen must be learned, not unsupervised (§5).
5. The finding that the gap is bound-limited, not compute-limited, with the implied next step (§6).

2 The N-track LR and its faithful reproduction

Model. Each train f has an origin, destination, entry time, speed multiplier, and intended arrival. The link run time is $\tau = \max\big(1, \lfloor \ell \cdot 60/(v \cdot s) + 1 \rfloor\big)$ for length ℓ, directional speed v, multiplier s. A traversal occupies link cells [enter, exit $+\, h$) for headway h; capacity is one train per (link, time) cell (zero during maintenance windows). The objective is the total absolute arrival deviation $\sum_f |\mathrm{arr}_f - \mathrm{intended}_f|$. Dwell is allowed only on siding links; departure carries a free slack window. Dualizing the per-cell capacity constraints with prices $\rho_{l,t} \geq 0$ decomposes the Lagrangian into per-train time-dependent shortest paths whose arc cost is the prefix-sum of ρ over the occupied window plus the destination-arc deviation. The bound is $\mathrm{LB} = \sum_f \min\text{-}\mathrm{cost}_f - \sum_{l,t} \rho_{l,t}$; a priority-rule heuristic gives a feasible upper bound.

Reproduction. On RAS Data Set 3 (85 nodes, 97 links, 40 trains, horizon 1440, headway 3), 10 subgradient iterations give the trajectory in Table 1. The bounds match the original MFC C++ (LB 1844 $\to$ 2359, UB $\sim$ 4153) and the published `summary.xlsx` (LB $\approx$ 1923, best UB $\approx$ 3471, gap $\approx$ 45%) in magnitude and dynamics; residual differences are the integer-bin "+1" run-time truncation and a single-pass priority heuristic.

Table 1: Faithful LR on RAS Set 3 (clean C++17, 10 subgradient iterations).

	free-flow	iter 1	iter 10 (best)
lower bound	1785	1785	**2279**
upper bound	—	4787	**3803**
gap	—	62.7%	**40.1%**

3 LR-based branch-and-bound

The LR alone cannot close the 40% gap; we wrap it in best-first B&B.

Node and bound. A node carries per-train time-windowed *restrictions* R (link l forbidden to train f during $[t_0, t_1)$), enforced inside the shortest path: a traversal whose occupancy window intersects a forbidden window is skipped. The node bound re-runs the subgradient LR under R, warm-started from the parent's multipliers, and returns $(\mathrm{LB}, \mathrm{UB}, \mathrm{primal})$. The root call reproduces Table 1 exactly, confirming the wrap is behavior-preserving.

Branching = sound disjunction + heuristic selection. We deliberately separate two concerns. *The disjunction* (sound, invariant) follows Zhou–Zhong (2005): pick a violated capacity cell (l, t^*) with conflicting trains i, j, and **fix one train's segment on l while the other yields**. The two children are the meet/pass *orders* (i-first vs j-first); on block-occupancy single track they are exhaustive (two trains on a unit-capacity link cannot overlap, so one whole occupancy window precedes the other), so the union of the children covers every feasible schedule and the search stays exact. A single-cell split (forbid one train from (l, t^*) only) is the always-valid fallback. *The selection* (heuristic, swappable) is *which* conflict to fix; it changes the tree's size, never correctness. We implement most-violated, earliest, latest, a dual score (§5), and a top-K dual-screened strong-branching oracle.

Exactness and node counts. On a hand-checkable native instance (3 trains, one corridor, headway 1, optimum = 12, arrivals $9, 13, 17$), the B&B exhausts the tree and proves the optimum (gap 0.00%). The segment enumeration proves it in **19** nodes versus **51** for the single-cell split ($\sim 2.7\times$ fewer), since each segment branch advances a whole occupancy window and lets the LR offset the rest. The selection criterion is also a real lever: under the cell split, the *latest*-conflict rule needs 19 nodes versus 51 for most-violated/earliest — "first thing first" is demonstrably not best, and any criterion keeps the proof valid.

Scaling. On RAS Set 3 the B&B closes the gap from the UB side: five nodes take the gap $40.1\% \to 31.2\%$ (LB $2279 \to 2348$, UB $3803 \to 3412$). Per-node cost (a full LR) is the bottleneck; §4 and the speed knobs below address it.

Per-node speed. Asymmetric LR depth (a converged root, few warm-started child iterations), computing the UB heuristic once per node rather than per iteration, and a wall-clock budget together give $\sim 8\times$ more nodes in fixed time (66 nodes / 220 s versus 20 / 540 s).

4 An exact, faster shortest path

Every arc of the time-expanded graph strictly increases time (run time ≥ 1), so the graph is a DAG and a single sweep in increasing time order computes exact shortest paths without a priority queue ($O(\mathrm{states} + \mathrm{arcs})$). This cuts the RAS root LR from 7.31 s to 2.48 s ($\sim 2.9\times$) at identical shortest-path *costs* (the free-flow baseline and the first two LR iterations match the Dijkstra bounds to the decimal across all 40 trains; the toy still proves optimum 12). The two implementations may pick different *equal-cost* paths on ties, so the subgradient trajectory and final root bounds differ slightly — standard LR degeneracy, both valid. We keep the classical Dijkstra as a verification fallback.

Remark 1 (Why not an approximate DP?)**.** *The LR lower bound is valid only if each per-train subproblem is solved to optimality (or to a valid lower bound). A greedy/heuristic path returns a* too-expensive *route, so* LB $=$ *trip* $- \sum \rho$ *becomes an over-estimate and B&B fathoming becomes unsound. Admissible accelerations are therefore exact (DAG DP, A***) or valid* relaxations *(coarser time grid rounded down; dropping dwell/headway) — never a greedy shortest path.*

5 Branch screening: primal, dual, SVD, tensor

At a B&P node of the space-time LP, a candidate action splits one fractional train's departure-time columns at a threshold θ (left = early, right = late). Full strong branching solves both child LPs for every candidate; we ask whether a *cheap* score (parent side, no child solve) contains the strong-branching choice.

Scores. For a column p let its dual exposure be $\sum_{c\in p}\rho_c$ and its primal exposure be $\sum_{c\in p}(\mathrm{H}x)_c$ (flow through its cells). The gradient-weighted scores are

$$\text{dual-gw}(a) = \text{frac}(a)\,\left|\overline{\rho\text{-load}}(\text{left}) - \overline{\rho\text{-load}}(\text{right})\right|,$$

and primal-gw identically with primal exposure. The *compressed-response* score projects each action's response signature b_a (frac-weighted left−right cell occupancy) onto a rank-r subspace of the response field and weights by the gradient g (ρ for dual, flow for primal): svd-cr$(a) = \left| b_a^\top U_r U_r^\top g \right|$, the realized form of $\sum_i \sigma_i(u_i^\top \nabla S)(v_i^\top b_a)$. The *tensor* score stacks the two channels into R[action, cell, {primal, dual}] and scores the gradient-weighted reconstruction from a Tucker (HOSVD) or CP-ALS low-rank fit, fusing primal and dual through a shared action/cell subspace.

Single root node: striking but fragile. On the RAS Set 3 LP root (20 candidates, 112 fractional, response-field rank-9 at 95% energy), the SVD scores excel (Table 2): **primal-SVD contains the strong-branching best exactly** (Containment@1= 1, rank gap 0) and **dual-SVD has the best rank correlation** (+0.70), while the raw gradient-weighted scores are weak (−0.07, +0.39) — compression is the enabler. Both contain the best within $K = 5$ (4× fewer child LP solves); pseudo-cost contains it in neither.

Table 2: Single root node (RAS Set 3 LP, 20 candidates). Striking — but see Table 3.

	Contain@1	Contain@5	Spearman
primal-gw	0	0	−0.07
dual-gw	0	0	+0.39
primal-SVD	**1**	**1**	+0.28
dual-SVD	0	**1**	**+0.70**
pseudo-cost	0	0	−0.64

Multiple nodes: it does not replicate. Walking six B&B nodes (descending by the strong-branching best, 48 candidate rows, 96 child solves) and adding the tensor scores gives Table 3. *No screening method robustly beats the baselines.* A random score already attains Containment@1= 0.33 and @5= 0.83 (eight-candidate sets make top-1 a one-in-eight shot), pseudo-cost wins Containment@1 (0.67, though with a negative Spearman: it pins top-1 but inverts the rest), and every SVD/tensor rank correlation is near zero. The striking single-node result was an $n=1$ artifact.

Table 3: Multiple B&B nodes (RAS Set 3, 6 nodes, 8 candidates each). The single-node signal vanishes.

	Contain@1	Contain@5	Spearman
primal-gw	0.33	0.83	+0.08
dual-gw	0.17	0.50	+0.08
primal-SVD	0.17	0.67	+0.02
dual-SVD	0.33	0.67	+0.12
tensor-HOSVD	0.17	0.67	−0.02
tensor-CP	0.17	0.83	+0.06
pseudo-cost	**0.67**	0.67	−0.47
random	0.33	0.83	+0.11

Interpretation. This is the same lesson the single-track study reached: an *unsupervised* spectral/compressed score is not a robust branch screen; only a *learned*, gradient-aware model transferred there (gradient boosting / reduced-rank regression contained the strong-branching action in the top-5 on 92% of nodes and beat pseudo-cost out-of-sample). The per-node response tensor is genuinely low multilinear rank (metric mode rank one), so the structure is real — but extracting a screening signal from it needs supervision, not an unsupervised projection. We also note low statistical power (six nodes, eight candidates): this is *weak evidence against*, not a clean refutation. The honest conclusion is that the robust N-track screen must be a learned transfer model over the primal+dual+tensor features, which is our next experiment.

6 Where the gap actually is

Using the $\sim 3\times$ faster path to afford more and deeper nodes (six nodes, dual selection, six warm child iterations) does *not* change the picture: the gap stays $\approx$36% and, crucially, **the global lower bound never lifts above the root LR value** ($\approx$2300); the gap closes only from the upper-bound side (incumbent $3803 \to 3561$). The reason is structural, not computational: the LR dual bound is inherently loose here (the published bound itself plateaus ≈ 2359 against a best UB ≈ 3471), and forbidding one train's segment per node barely raises that node's bound, so best-first keeps a frontier pinned near the root. *Faster nodes buy more nodes, not a tighter bound.* The implied levers are therefore on the bound: a crossing-conflict / enhanced lower bound (which cut nodes $7.5\% \to 35.6\%$ on single track), branching that fixes a bottleneck resource's full ordering at once, or a tighter relaxation entirely — the cumulative-flow LP master or an ADMM consensus bound that iteratively fixes train variables. Branch screening helps the selection/UB side; the lower-bound side needs a stronger relaxation.

7 Conclusions and next steps

We delivered an exact, portable, reproducible N-track LR-based B&B with segment-enumeration branching and a $\sim 3\times$ faster exact shortest path, and we evaluated primal/dual/SVD/ tensor branch screening without spin: a single-node success that did not survive a multi-node test, consistent with the single-track finding that only a *learned* screen transfers. The optimality gap is bound-limited, not compute-limited. Two next steps follow directly: (1) a learned transfer screen over the primal+dual+tensor features, trained on some nodes/instances and tested out-of-sample against strong branching; and (2) a stronger per-node relaxation (cuts or an ADMM consensus bound) to lift the global lower bound, which is the only thing that will close the gap.

Reproducibility. All results come from the handoff package (`train_scheduling/FTT_handoff`): the C++ LR/B&B (`n_track/fasttrain.cpp`; build `g++ -O2 -std=c++17 -static`; run `--bnb [--split segment|cell] [--branch most-violated|earliest|latest|dual|strong] [--sp dag|dijkstra] [--child-iters N] [--time S]`), the LP/screening (`nt_spacetime.py`, `nt_screen.py`), the multi-node tensor screening (`nt_tensor.py`), and the data (`data/RAS_set3_native`, `data/toy_native`, `data/arc_format_*`). The single-track engine and screening live in `single_track/`.

References. Meng, L. and Zhou, X. (2014), *Simultaneous train rerouting and rescheduling on an N-track network: A model reformulation with network-based cumulative flow variables*, Transportation Research Part B. Zhou, X. and Zhong, M. (2005), *Bicriteria train scheduling for high-speed passenger railroad planning applications*, European Journal of Operational Research. Zhou, X. and Zhong, M. (2007), *Single-track train timetabling with guaranteed optimality: Branch-and-bound algorithms with enhanced lower bounds*, Transportation Research Part B.